MEDIXO EDUX / EXAMINATION INTELLIGENCE BRIEFING

EXAM INTELLIGENCE REPORT

QA Three-Subject Fixture — extensive revision priorities and mathematical solutions

JEE Main 2027 · Full Syllabus Mock · JEE Main (Paper 1) · All Subjects · Generated 02 Nov 2026

STUDENT	ROLL NUMBER	EXAMINATION DATE	DURATION
Aarav Sharma	21CS114	2026-08-01	3 hrs

MARKS OBTAINED

244 / 360

QUESTION ACCURACY

72.6%

244 / 360 marks · 61 correct | 23 incorrect | 6 skipped · Attempted 84 / 90 · Time 161m

Grade A · Rank 1260 · Percentile 94.2 · Readiness 78

Subject performance

PHYSICS	CHEMISTRY	MATHEMATICS
88 / 120 · 76% · Rank 6 · Medium	80 / 120 · 71% · Rank 9 · Easy	76 / 120 · 63% · Rank 14 · Hard

Chapter mastery

Chapter mastery by subject

CHAPTER	ACCURACY	MASTERY
Physics		
Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	44%	Strong
Electrostatics	51%	Strong
Modern Physics	58%	Strong
Ray Optics	65%	Weak
Chemistry		
Physical Chemistry	72%	Strong
Organic Chemistry	79%	Average
Inorganic Chemistry	86%	Average
Coordination Compounds	45%	Weak

Chapter mastery by subject · continued

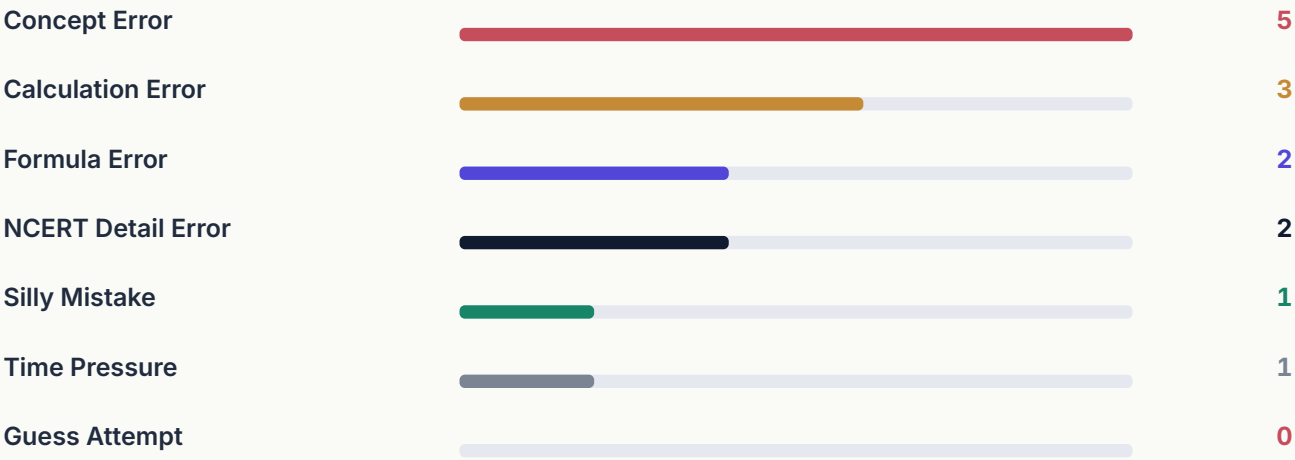
CHAPTER	ACCURACY	MASTERY
Mathematics		
Algebra	52%	Strong
Coordinate Geometry	59%	Average
Calculus	66%	Critical
Probability & Stats	73%	Critical

02 LEARNING INTELLIGENCE

Error patterns and topic accuracy

Mistake classifications, topic-level accuracy, subject-wise weak topics and source evidence.

Mistake pattern analysis



Topic-level accuracy

TOPIC	ACCURACY	LEVEL
Physics		
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	38%	Critical
Work-Energy-Power	47%	Weak
Electrostatics — Gauss Law	56%	Average
Thermodynamics (Physics)	65%	Strong
Ray Optics — Lenses	74%	Critical
Semiconductors	83%	Weak

Topic-level accuracy · continued

TOPIC	ACCURACY	LEVEL
Chemistry		
Mole Concept	92%	Average
Chemical Bonding	43%	Strong
Coordination Compounds	52%	Critical
Mathematics		
Definite Integration	61%	Weak
3D Geometry	70%	Average
Probability	79%	Strong
Quadratic Equations	88%	Critical
Matrices & Determinants	39%	Weak

Subject-wise weak topics

SUBJECT	WEAK TOPIC
Physics	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions
Mathematics	Definite Integration
Mathematics	3D Geometry
Physics	Ray Optics sign convention
Chemistry	Coordination Compounds — NCERT tables
Mathematics	Probability — Bayes & distributions
—	Thermodynamic non-ideal corrections
—	Electrostatic boundary conditions

Source mistake evidence

QUESTION	SOURCE DETAIL
Q14	Misapplied substitution — used u = x^2 instead of u = x^2+1
Q23	Decimal error in $1/r^2$ computation
Q31	Confused magnetic moment values for d^4 vs d^5

Priority classification / published list

PRIORITY TOPIC	CLASS	WHEN
Critical priority topic 1 — Definite Integration — properties & substitution	Critical	48 hours
High priority topic 2 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	5 days
Medium priority topic 3 — Definite Integration — properties & substitution	Medium	7 days

Priority classification / published list · continued

PRIORITY TOPIC	CLASS	WHEN
Critical priority topic 4 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Critical	10 days
High priority topic 5 — Definite Integration — properties & substitution	High	48 hours
Medium priority topic 6 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Medium	5 days
Critical priority topic 7 — Definite Integration — properties & substitution	Critical	7 days
High priority topic 8 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	10 days
Medium priority topic 9 — Definite Integration — properties & substitution	Medium	48 hours
Critical priority topic 10 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Critical	5 days
High priority topic 11 — Definite Integration — properties & substitution	High	7 days
Medium priority topic 12 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Medium	10 days

03 REVISION STRATEGY

What to study next

Published priority sequence, classification, subject-wise revision sections and timeframe.

Recommended study sequence

Order reproduced from the published priority revision list.

Priority board · most actionable page

SEQ.	PRIORITY TOPIC	CLASSIFICATION	REVISION TIMEFRAME
01	Critical priority topic 1 — Definite Integration — properties & substitution	Critical	48 hours
02	High priority topic 2 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	5 days
03	Medium priority topic 3 — Definite Integration — properties & substitution	Medium	7 days
04	Critical priority topic 4 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Critical	10 days
05	High priority topic 5 — Definite Integration — properties & substitution	High	48 hours

Priority board · most actionable page · continued

SEQ.	PRIORITY TOPIC	CLASSIFICATION	REVISION TIMEFRAME
06	Medium priority topic 6 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Medium	5 days
07	Critical priority topic 7 — Definite Integration — properties & substitution	Critical	7 days
08	High priority topic 8 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	10 days
09	Medium priority topic 9 — Definite Integration — properties & substitution	Medium	48 hours
10	Critical priority topic 10 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Critical	5 days
11	High priority topic 11 — Definite Integration — properties & substitution	High	7 days
12	Medium priority topic 12 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Medium	10 days

CRITICAL	HIGH	MEDIUM
4	4	4

Subject-wise revision sections

Source revision groups ordered by the analysis engine.

Priority topics grouped by subject

PRIORITY TOPIC	CHAPTER	ACCURACY	CLASSIFICATION
Physics			
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	—	38%	Critical
Work-Energy-Power	—	47%	Weak
Electrostatics — Gauss Law	—	56%	Weak
Mathematics			
Matrices & Determinants	—	39%	Critical
Definite Integration	—	61%	Review
3D Geometry	—	—	Revise
Chemistry			
Chemical Bonding	—	43%	Critical

Priority topics grouped by subject · continued

PRIORITY TOPIC	CHAPTER	ACCURACY	CLASSIFICATION
Coordination Compounds	—	52%	Weak

Also revise

Ray Optics sign convention	—	—	Revise
Coordination Compounds — NCERT tables	—	—	Revise
Probability — Bayes & distributions	—	—	Revise
Thermodynamic non-ideal corrections	—	—	Revise
Electrostatic boundary conditions	—	—	Revise

Published next actions

Practice, PYQ, mock and lecture resources attached to this attempt.

Recommended action register

ACTION	RESOURCE	DETAIL
PYQ	JEE Main 2024 — Definite Integration (5 Qs)	PYQ Bank
PYQ	JEE Main 2023 — Ray Optics (4 Qs)	PYQ Bank
PYQ	JEE Advanced 2022 — Coordination Compounds	PYQ Bank
Practice	20-question drill — Integration by substitution	20 · Medium
Practice	10-question drill — Lens formula applications	10 · Medium
Mock test	Sectional mock — Mathematics (60 min)	Sectional
Mock test	Full syllabus mock — JEE Main pattern	Full
Lecture	Integral Calculus — Masterclass (45 min)	Video
Lecture	Ray Optics — Sign Convention Simplified	Video

Source mistake evidence / detailed register

Q	SUBJECT	CATEGORY	SOURCE DETAIL
Q14	Mathematics	Concept Error	Misapplied substitution — used u = x^2 instead of u = x^2+1
Q23	Physics	Calculation Error	Decimal error in 1/r^2 computation
Q31	Chemistry	NCERT Detail Error	Confused magnetic moment values for d^4 vs d^5
Q47	Mathematics	Formula Error	Wrong distance formula between skew lines
Q52	Physics	Concept Error	Sign convention error in lens formula
Q63	Mathematics	Silly Mistake	Counted 36 outcomes instead of 36 — swapped total
Q71	Chemistry	Time Pressure	Rushed mechanism — skipped a step